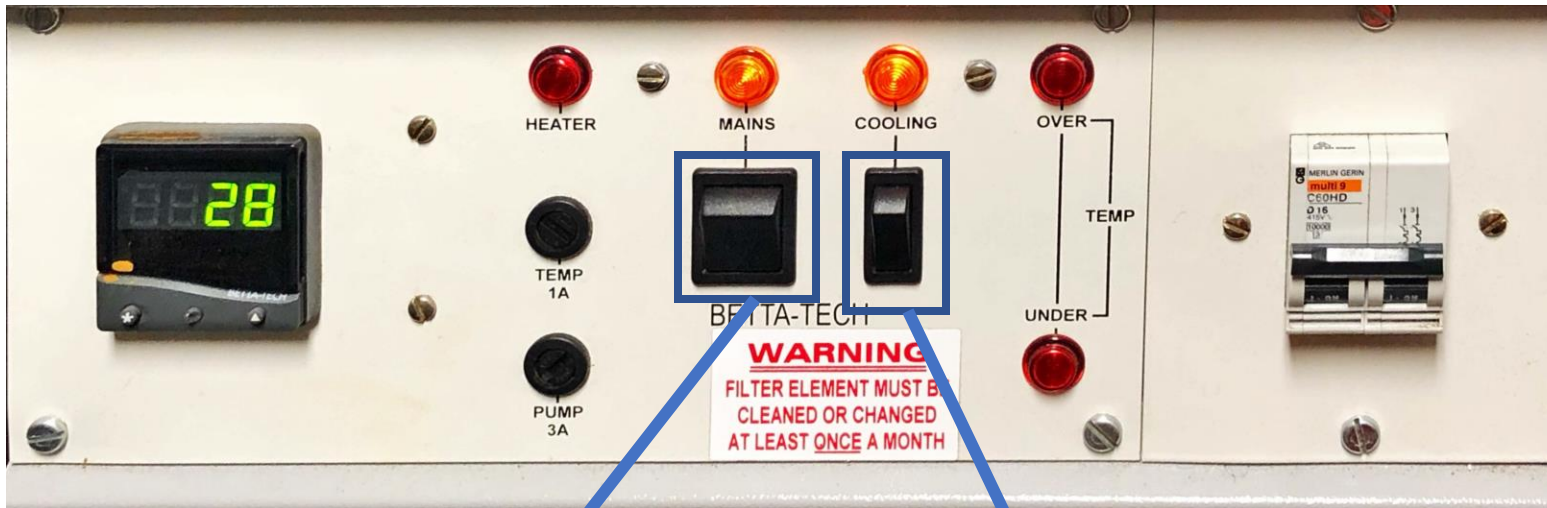


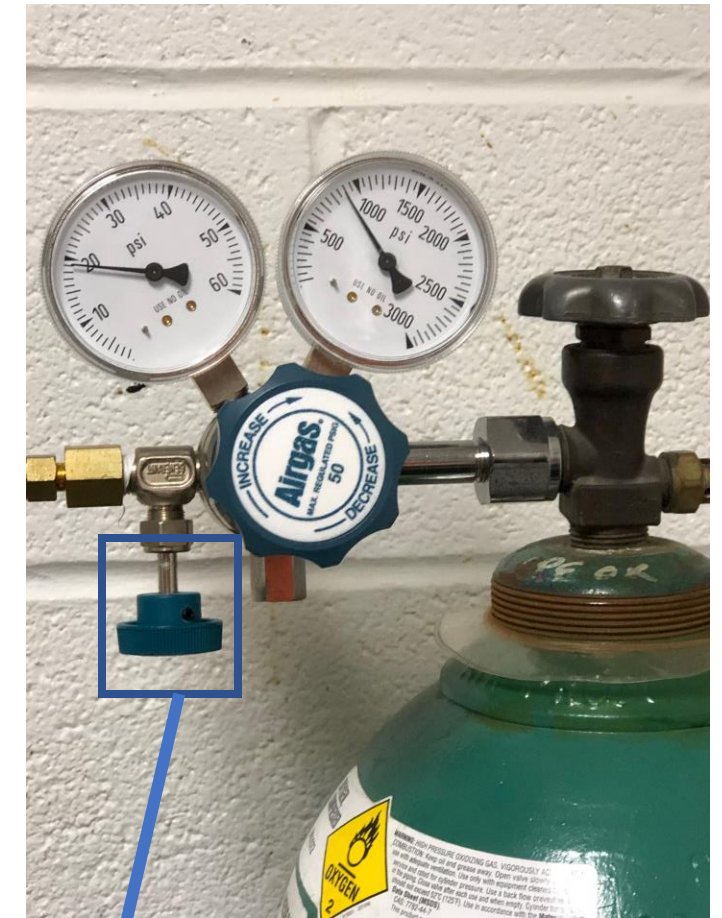
RIE Operational Manual

Back room



1. turn on "MAINS"

2. turn on "COOLING"

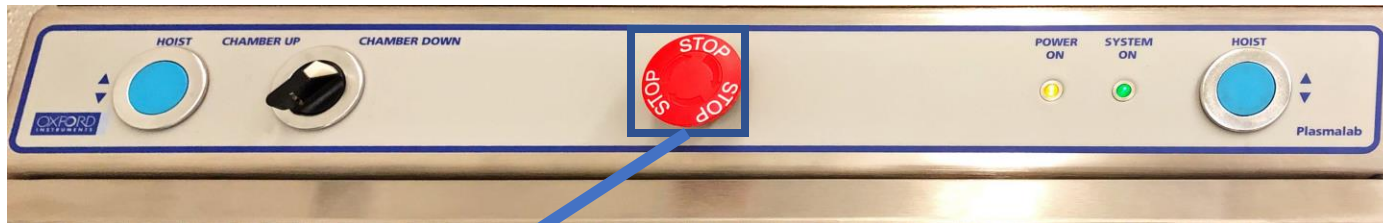


3. rotate the valve to left



4. turn on both
Compressor
and N2 Valves

6. Push green "ON" button

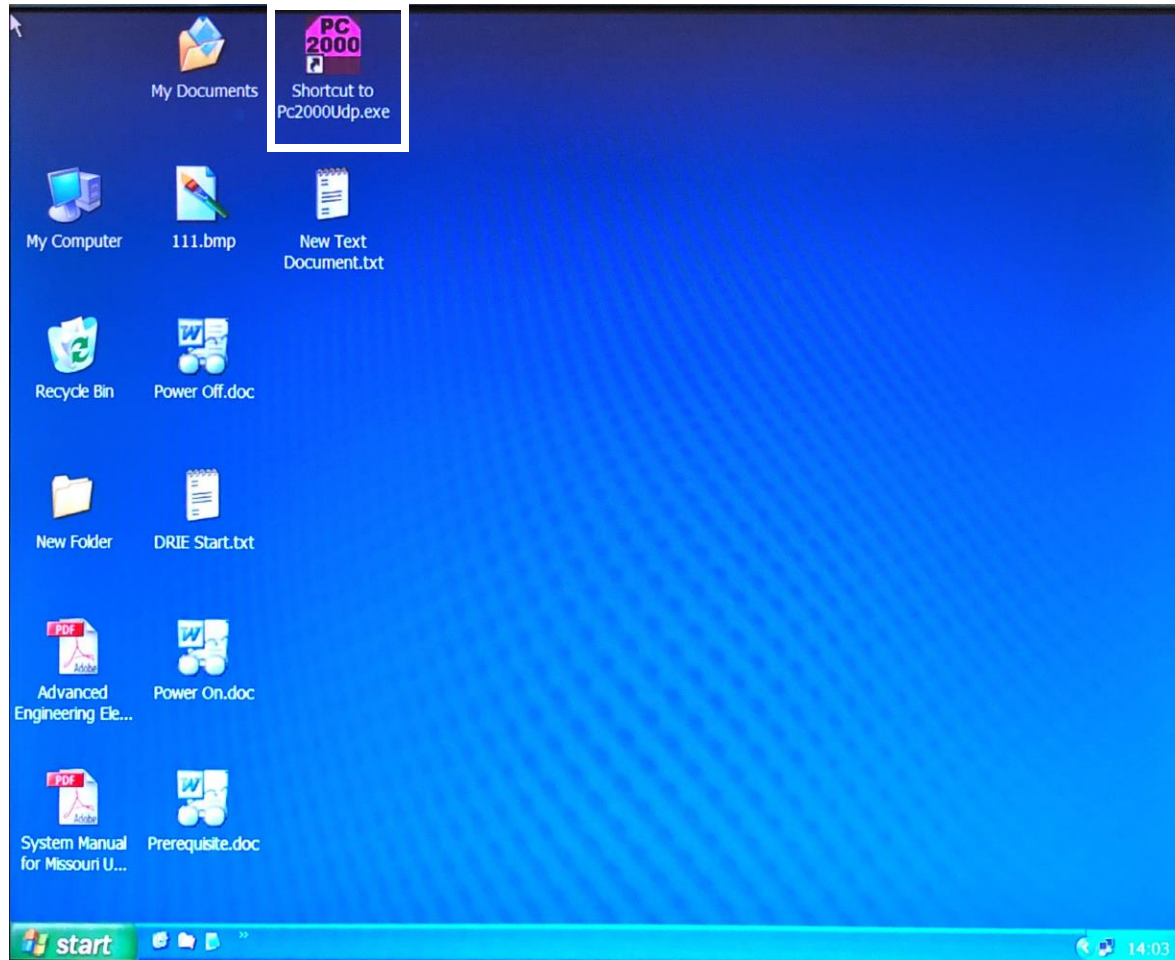


5. Clockwise to
release "STOP" button

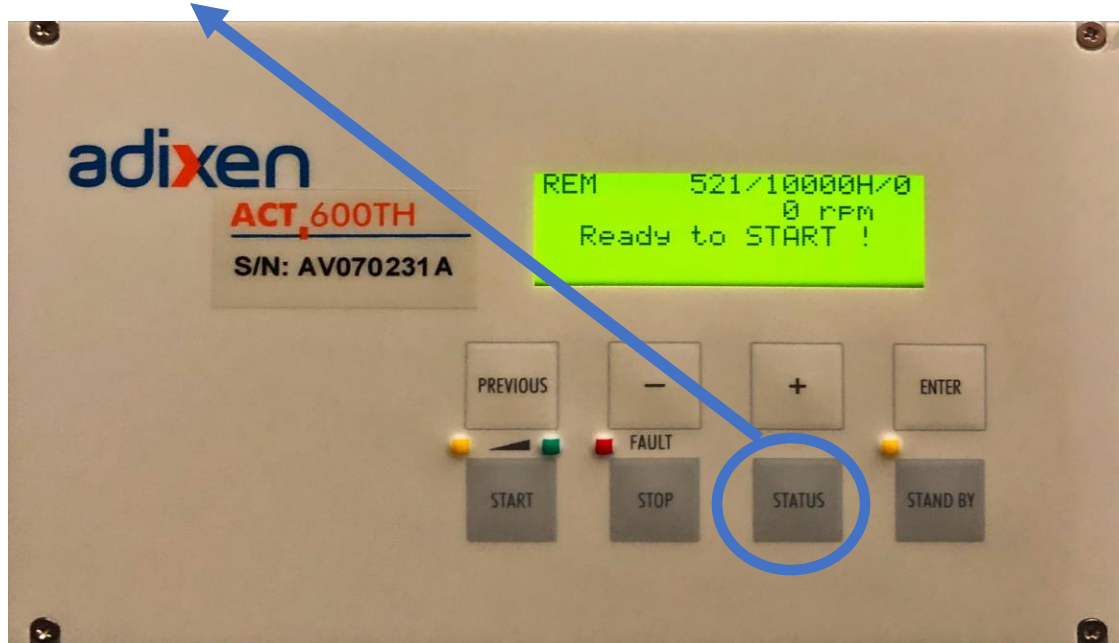
7. turn on small chiller,
wait for 20 °C



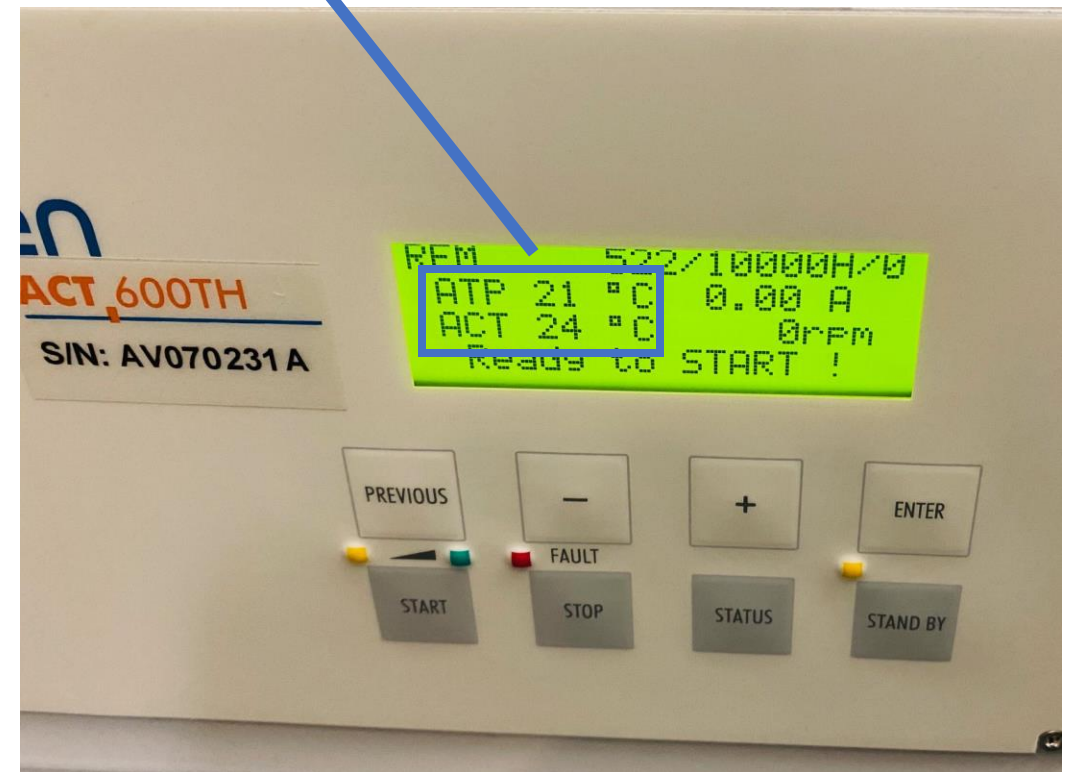
8. open software, Name: OPT
Password: OPT
9. ACCEPT all alerts



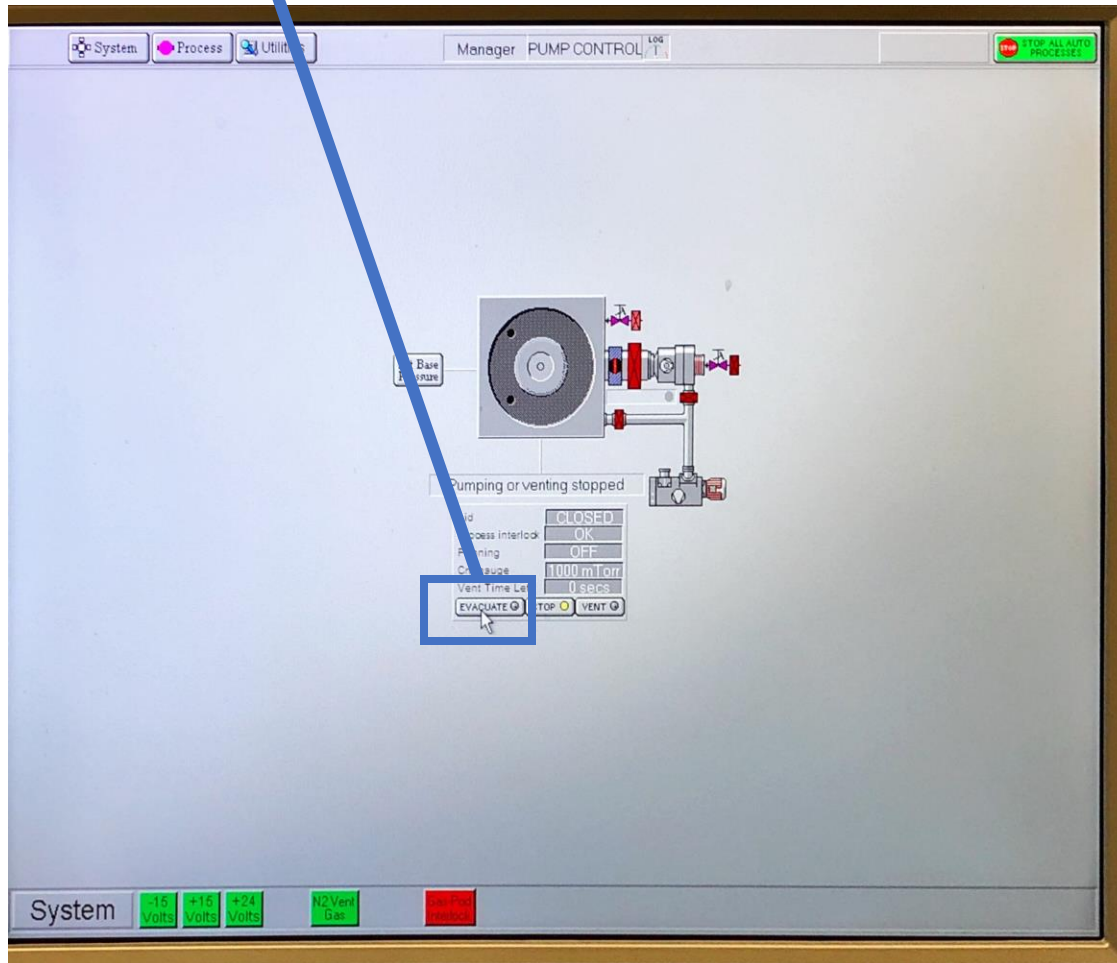
10. Press the status button



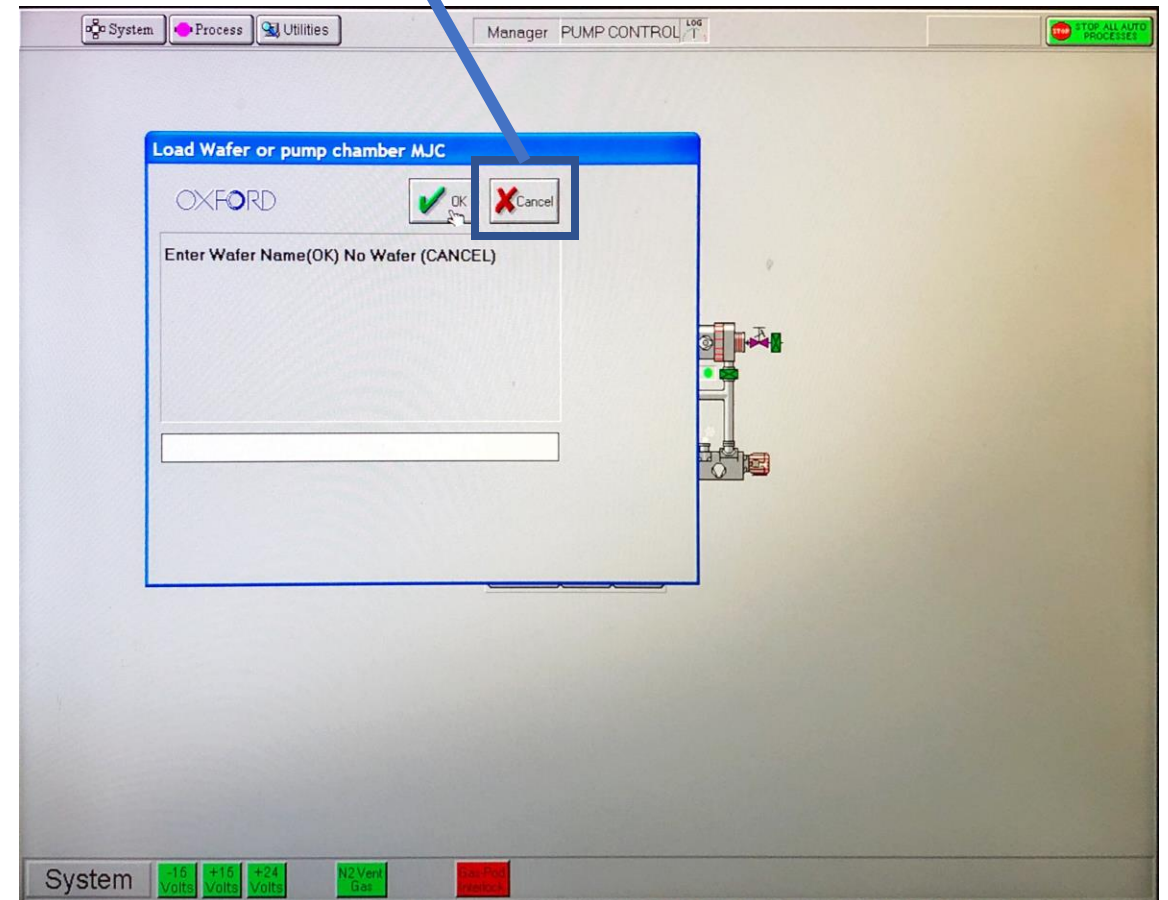
11. ATP is the temp of turbo pump
ACT is the temp of the machine controller
KEEP AN EYE ON THEM ALL THE TIME



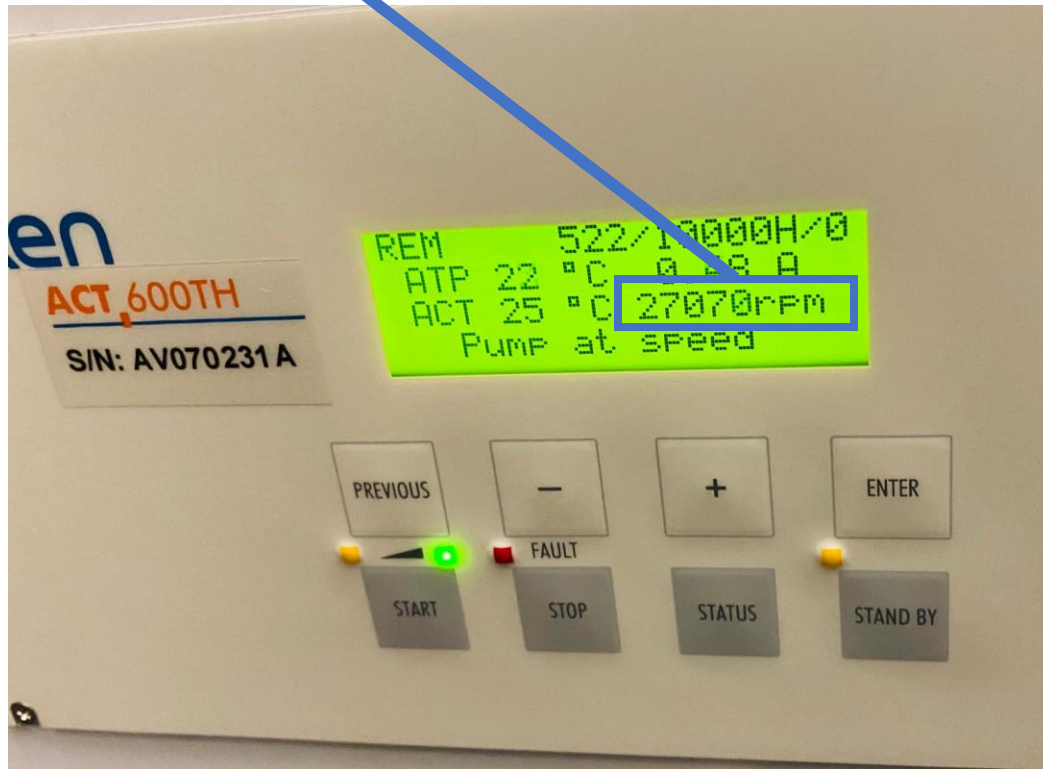
12. Click "Evacuate"



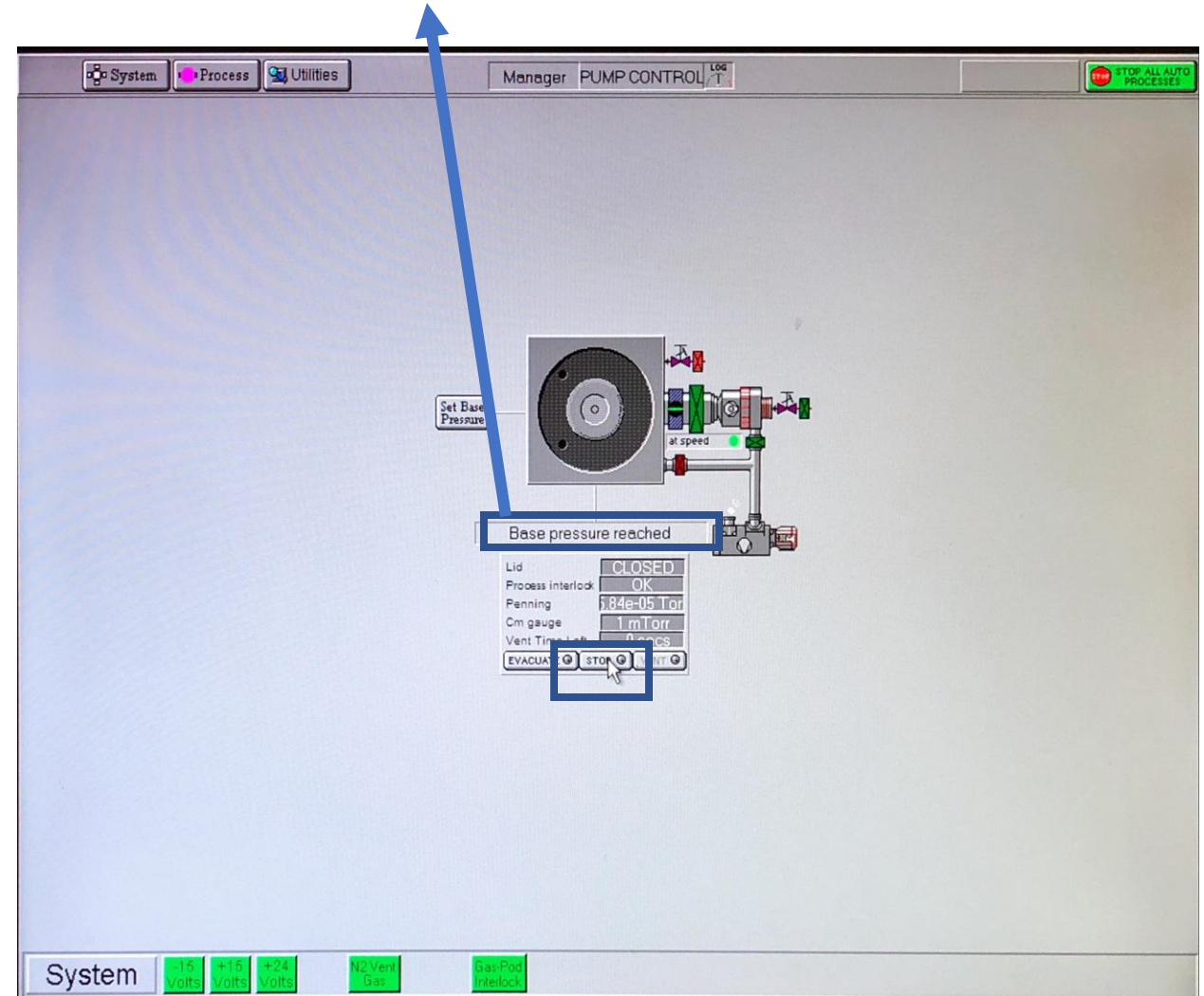
13. Click "Cancel" as there is no wafer in it



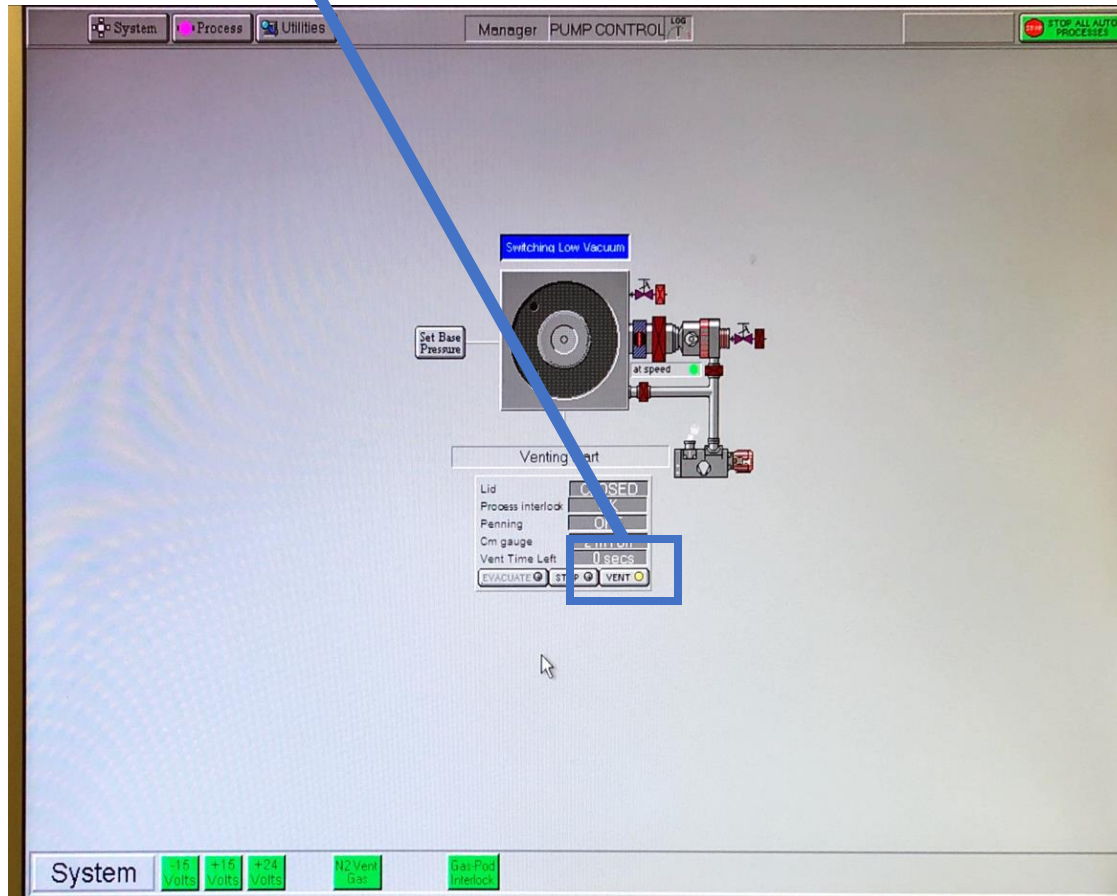
14. wait for turbo to reach 27000 rpm



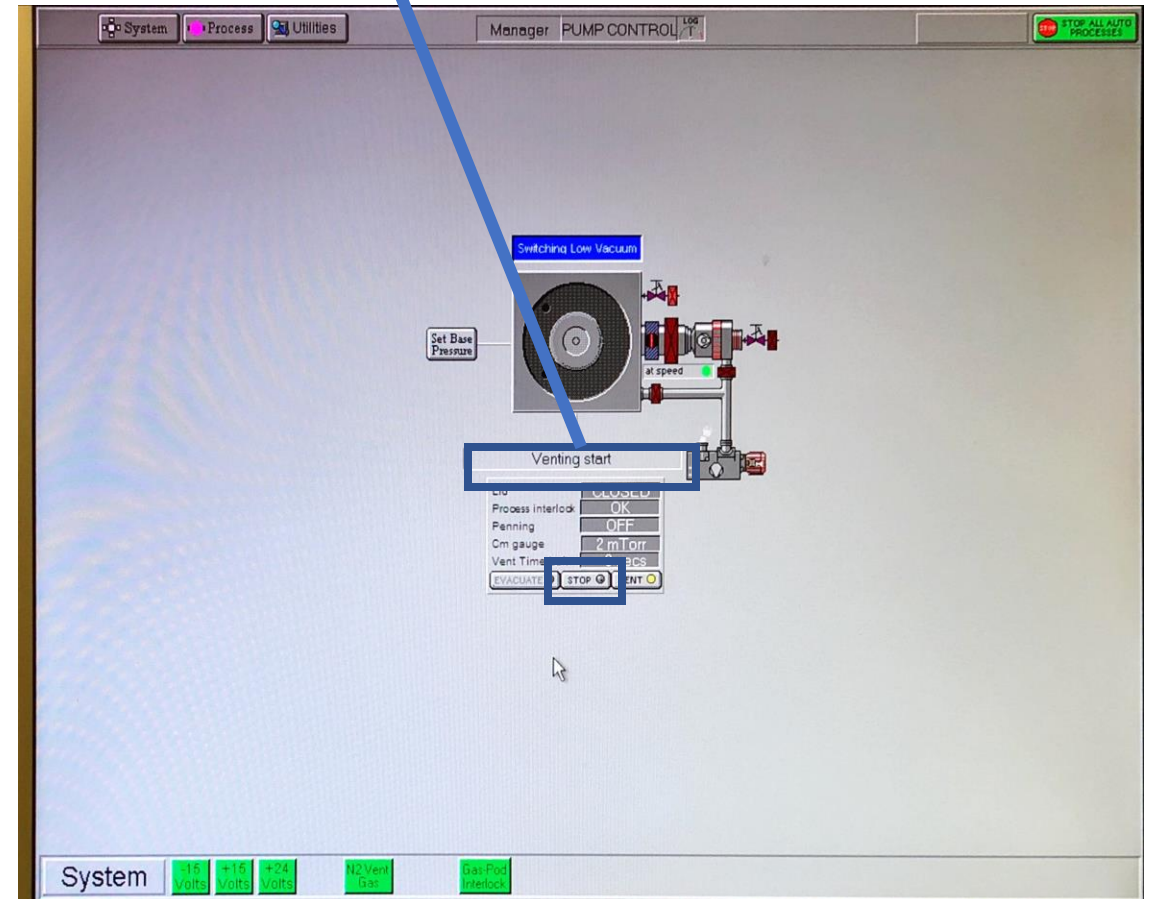
13. After seeing "Base pressure reached", STOP the evacuation



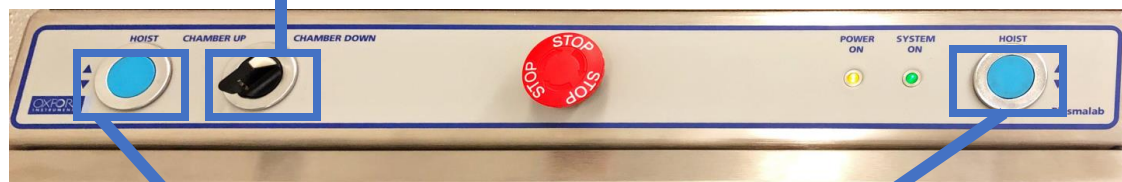
14. Click "Venting"



15. "Stop" venting when the status shows "Finished"

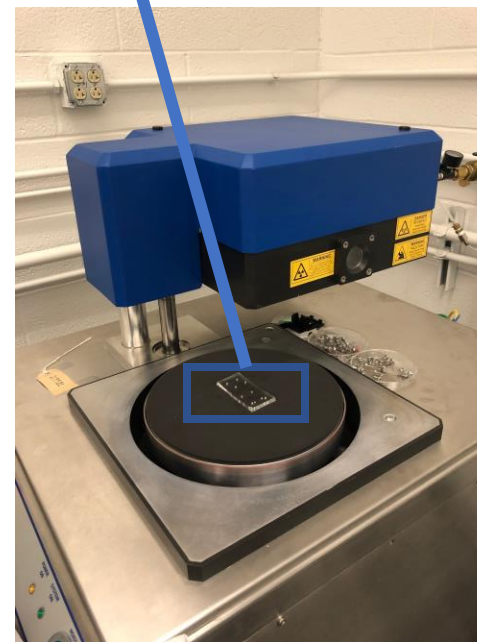


16. switch to "CHAMBER UP"



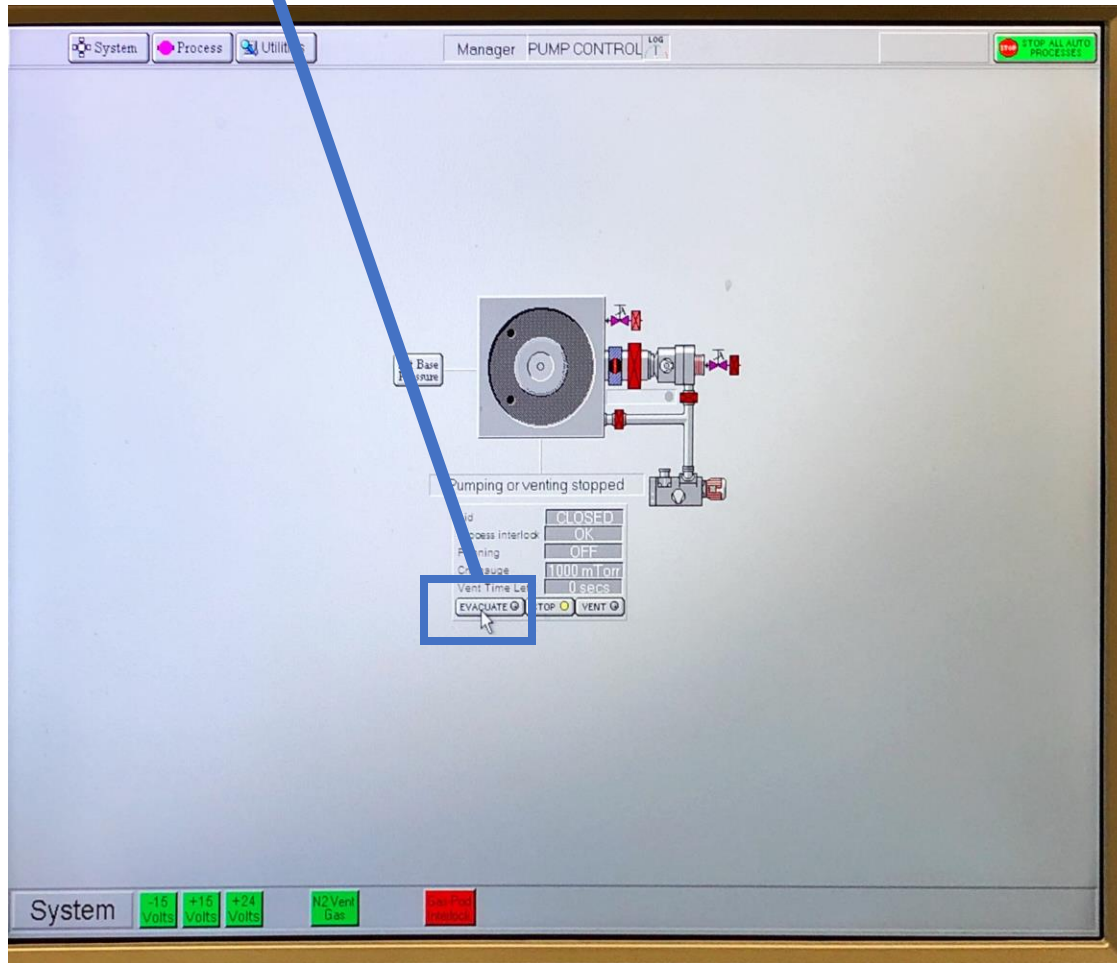
17. Press both buttons together until the lid completely opens up

18. Put sample on the stage (tape it if necessary)

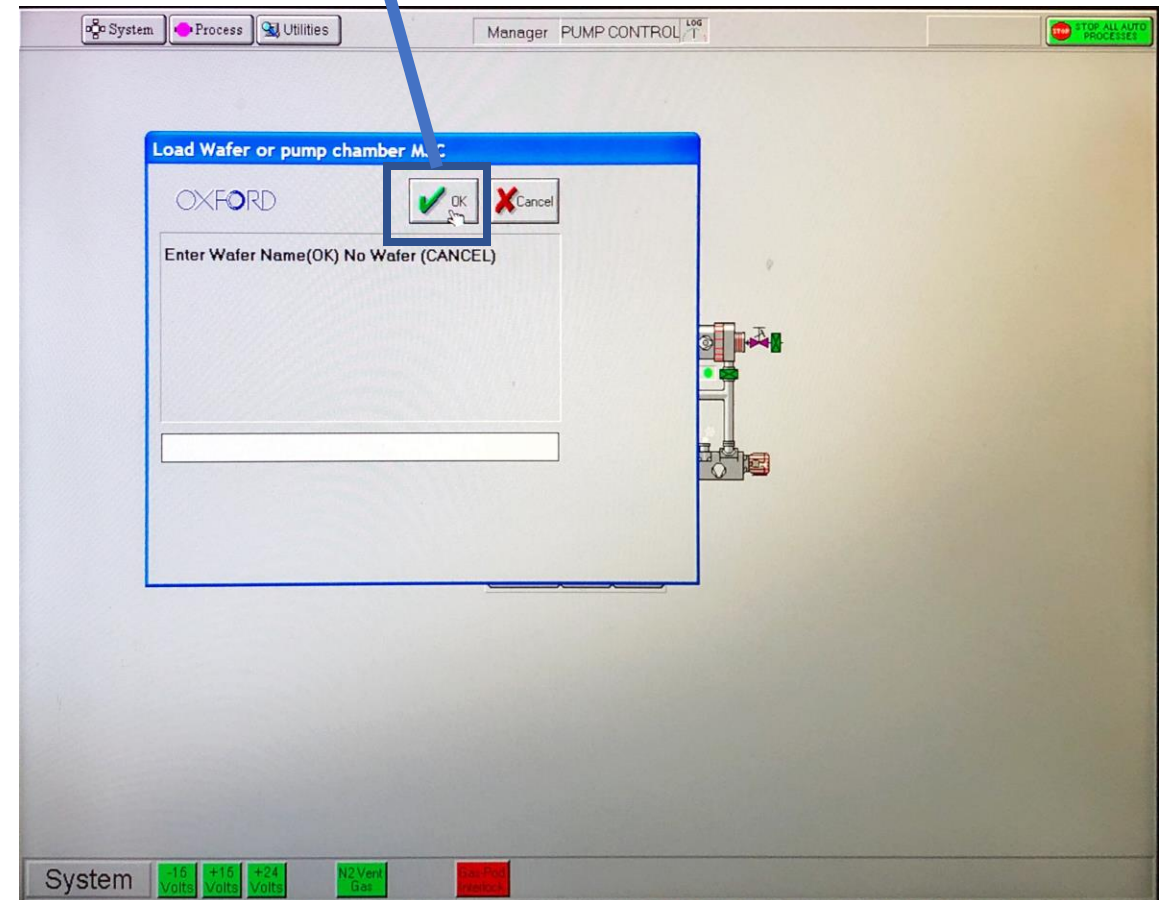


19. Turn the knob to "CHAMBER DOWN" and press both the hoist together until the lid completely closes

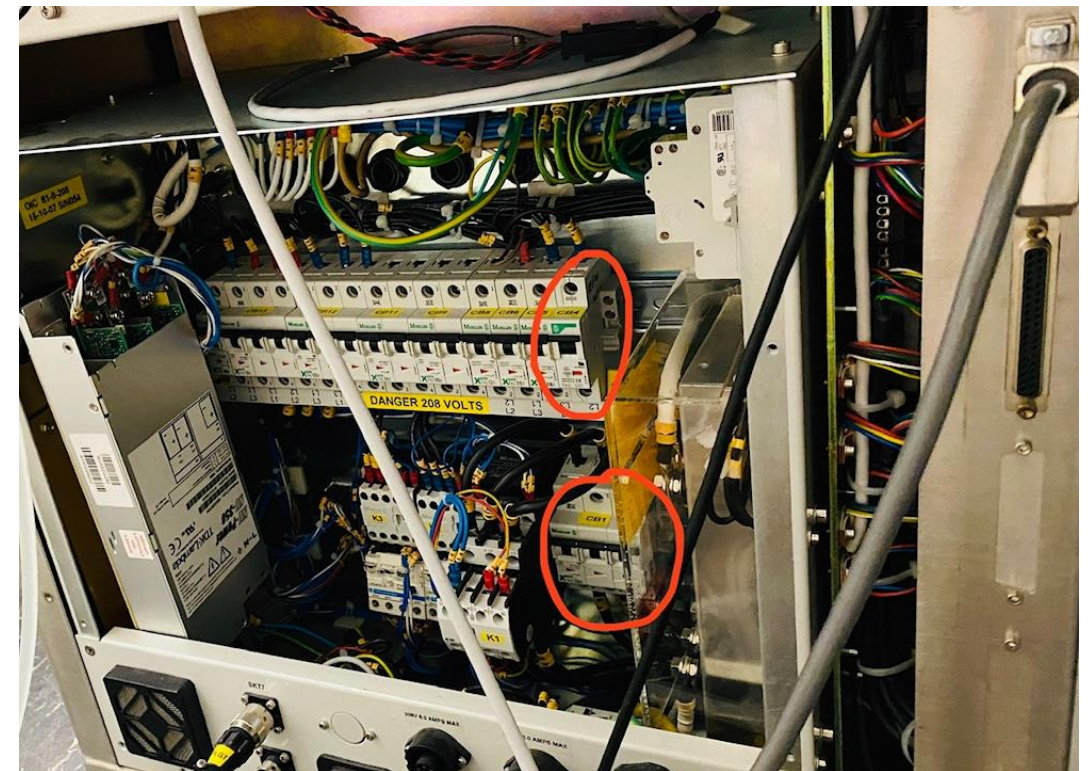
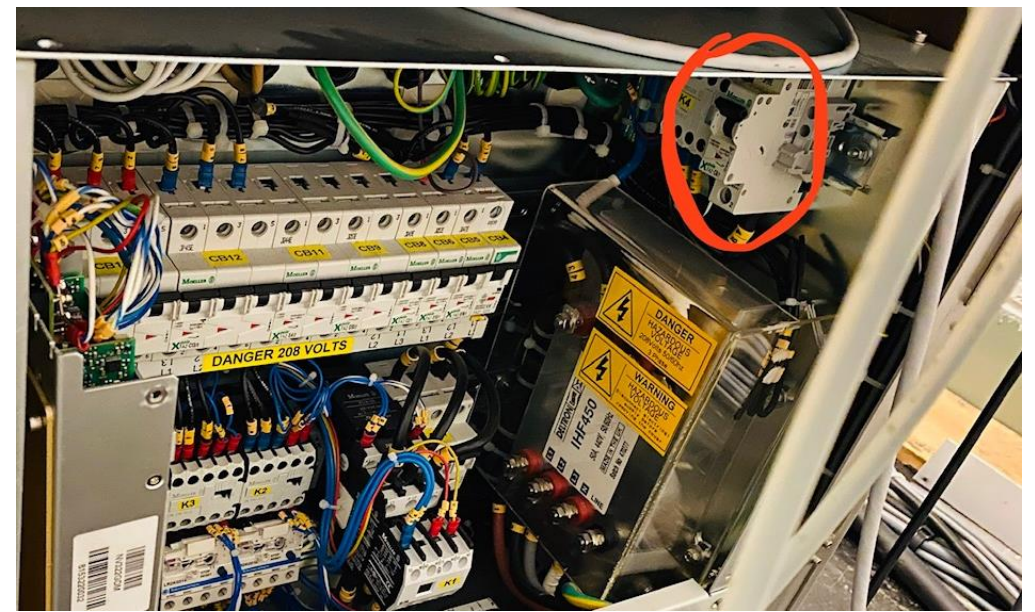
20. Click "Evacuate"



21. Click "OK" as sample is in it and wait for the screen to show "BASE PRESSURE REACHED"



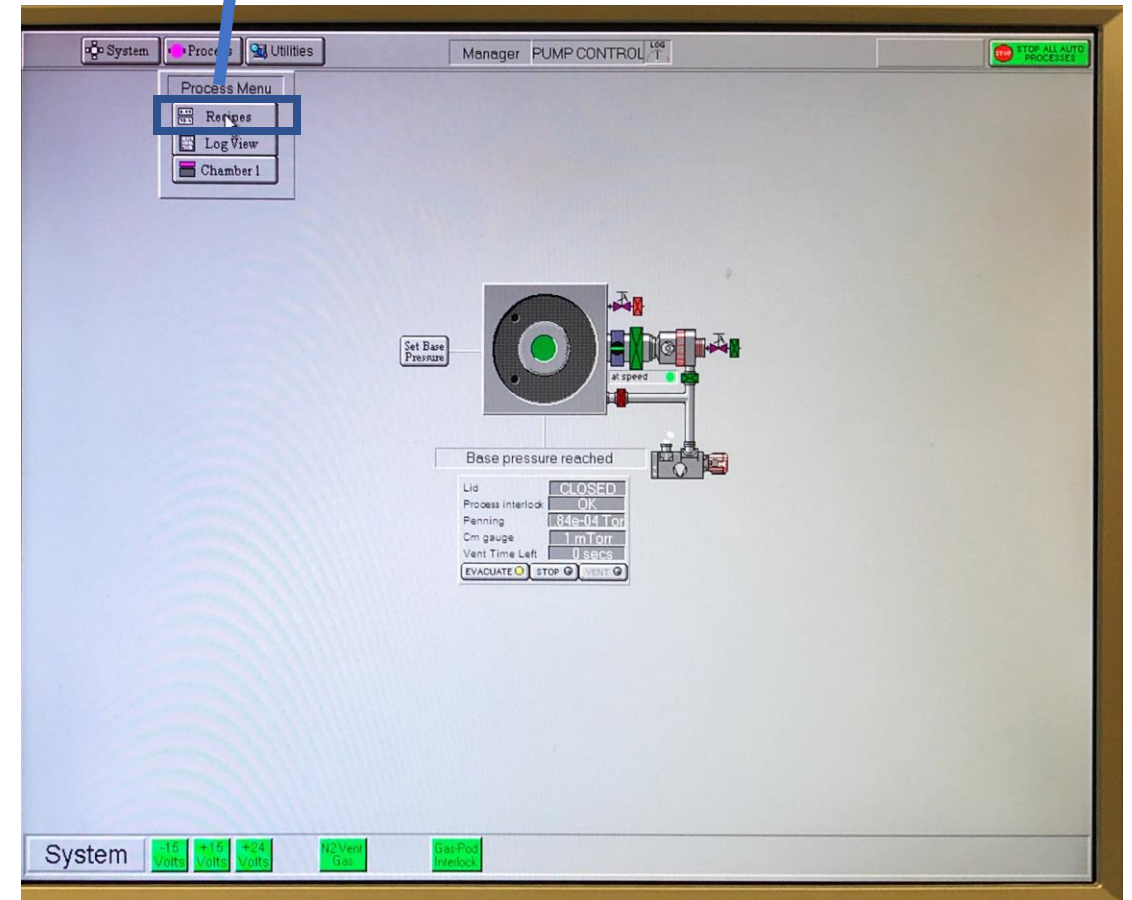
22. turn on "AC LINE" SLOWLY and make sure the green light is on.
23. If the green light doesn't turn ON, DO NOT PLAY WITH THE AC LINE, go back of the RIE and one of the circuit breakers have been tripped.
24. Turn the breaker back ON



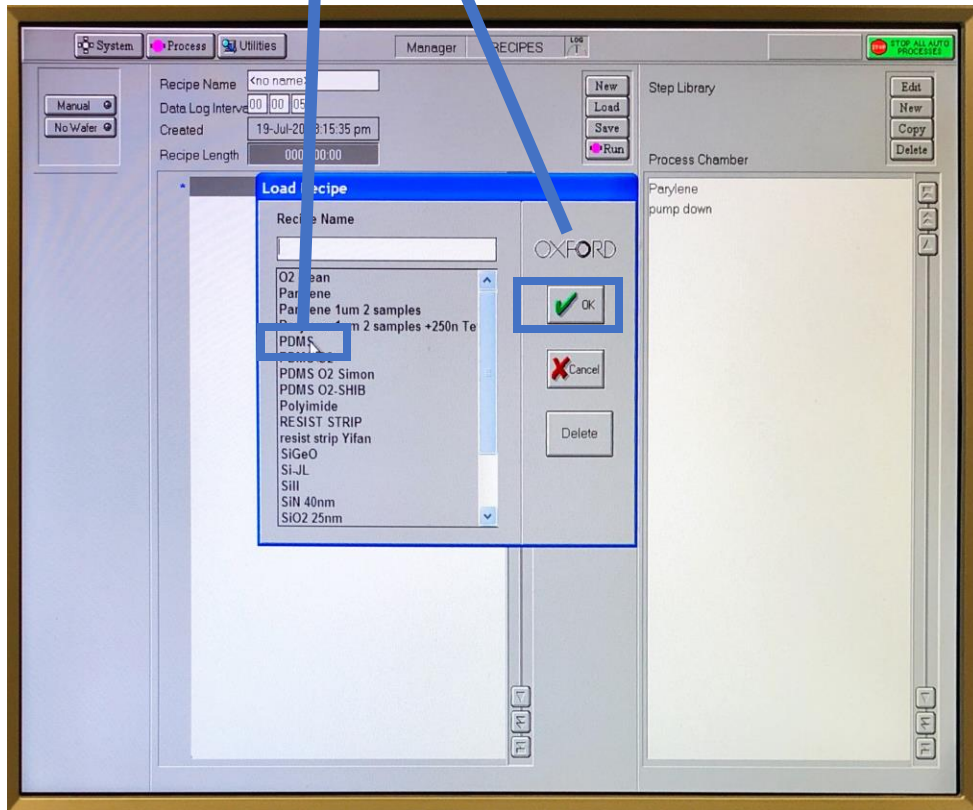
25. Make sure the AC ON is glowing now



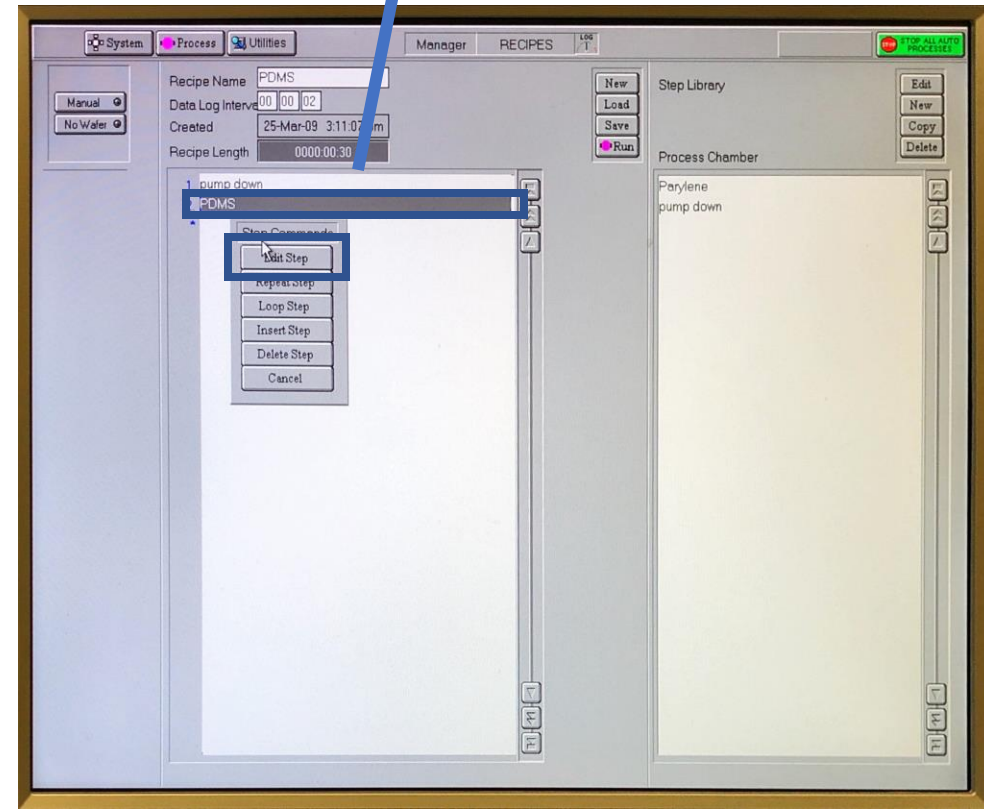
26. go to "Process" - "Recipe"



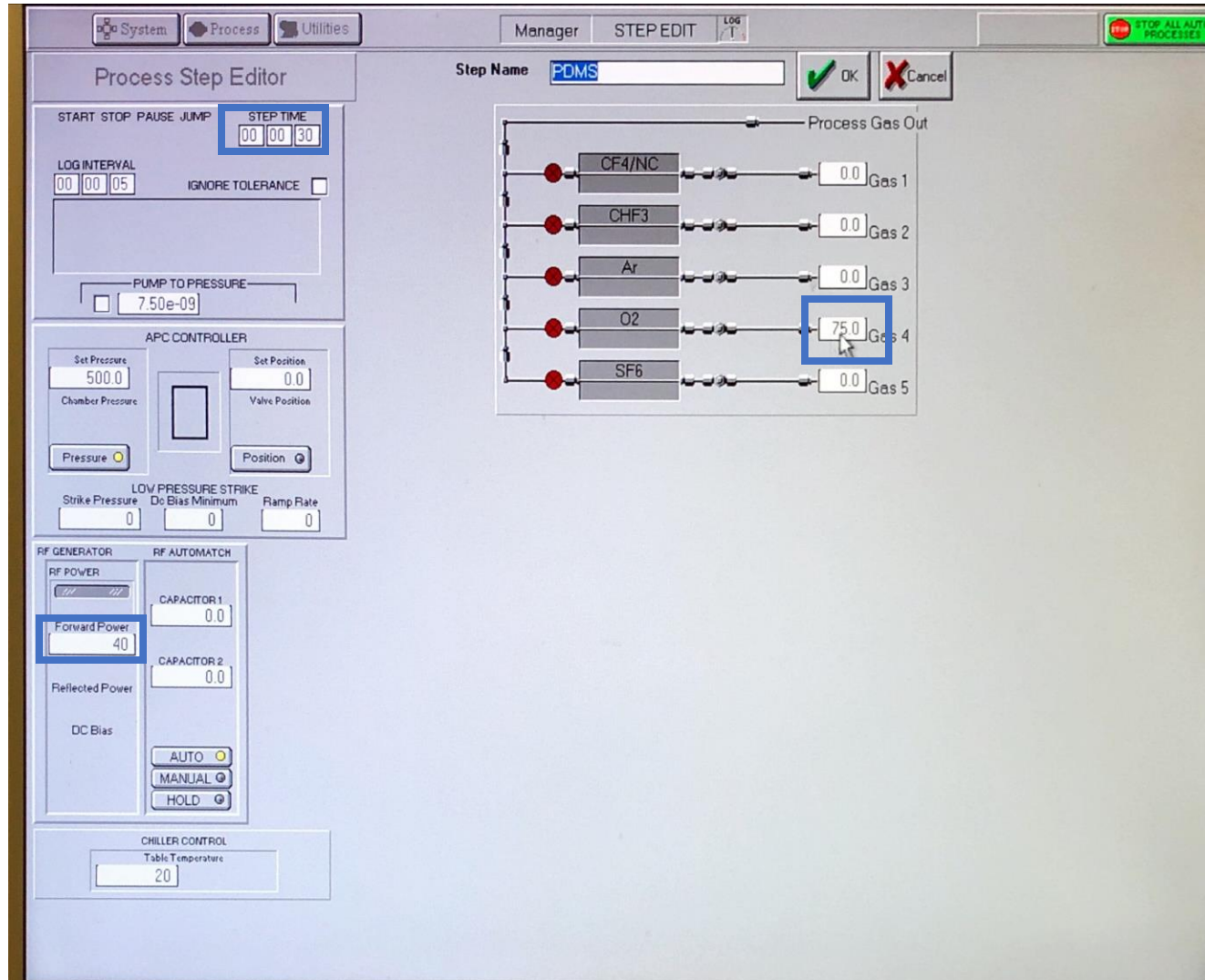
27. Click "LOAD" and then select "PDMS" (or any recipe that you want) and "OK"



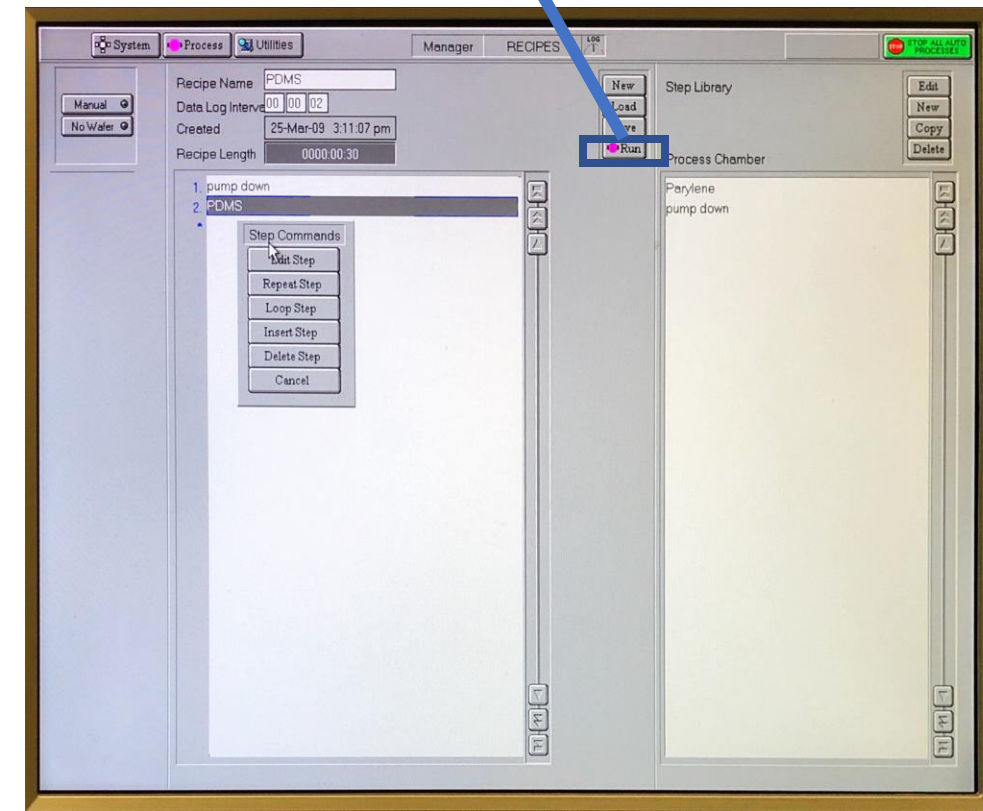
28. Click the recipe and then click "Edit Step"



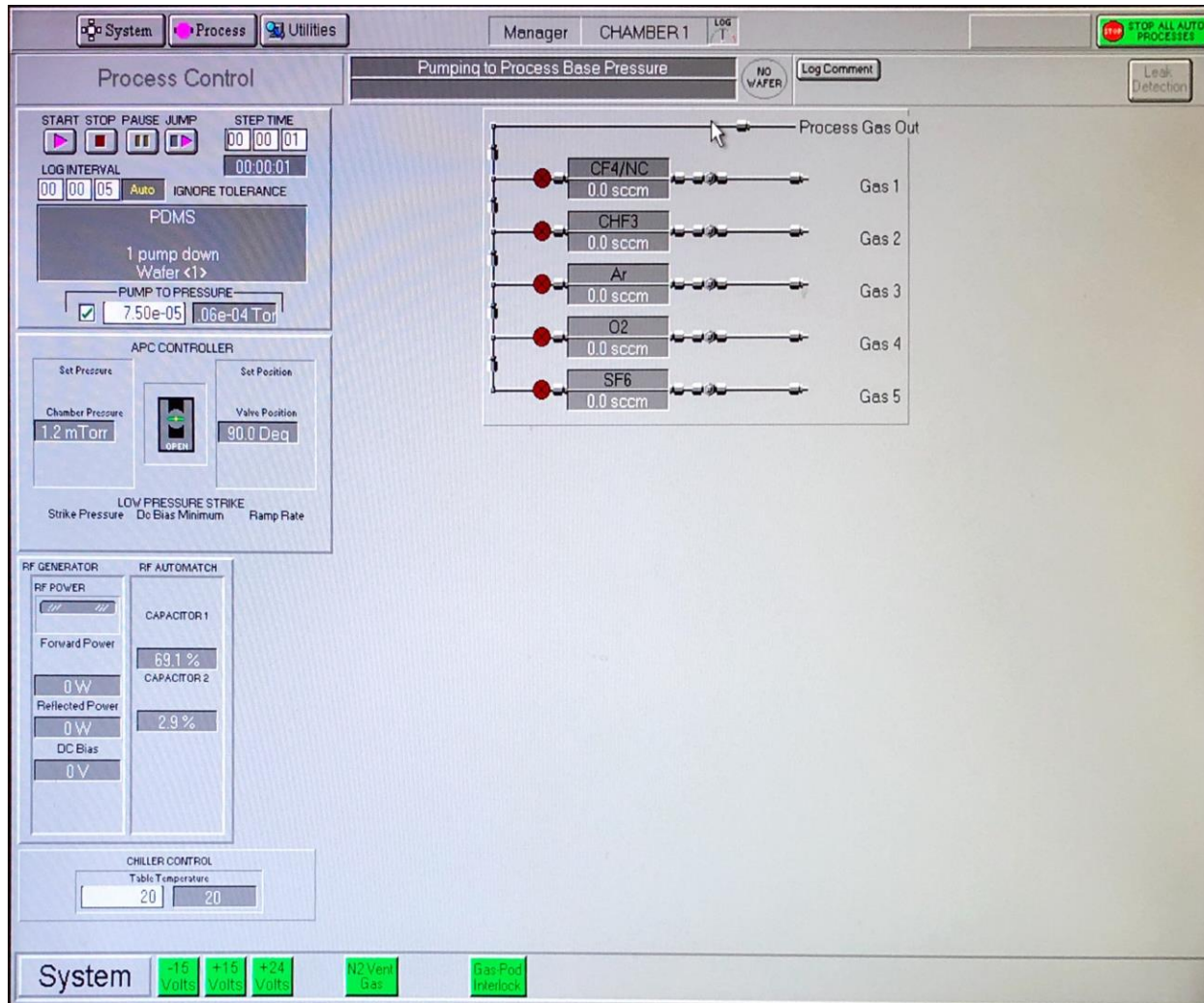
29. make sure the flow rate, process time, power numbers are right



30. "Run" the recipe



31. check for “all tolerance ok” and look for plasma in the chamber
32. While the process is running, keep an eye for flow rate, forward power and plasma in the chamber

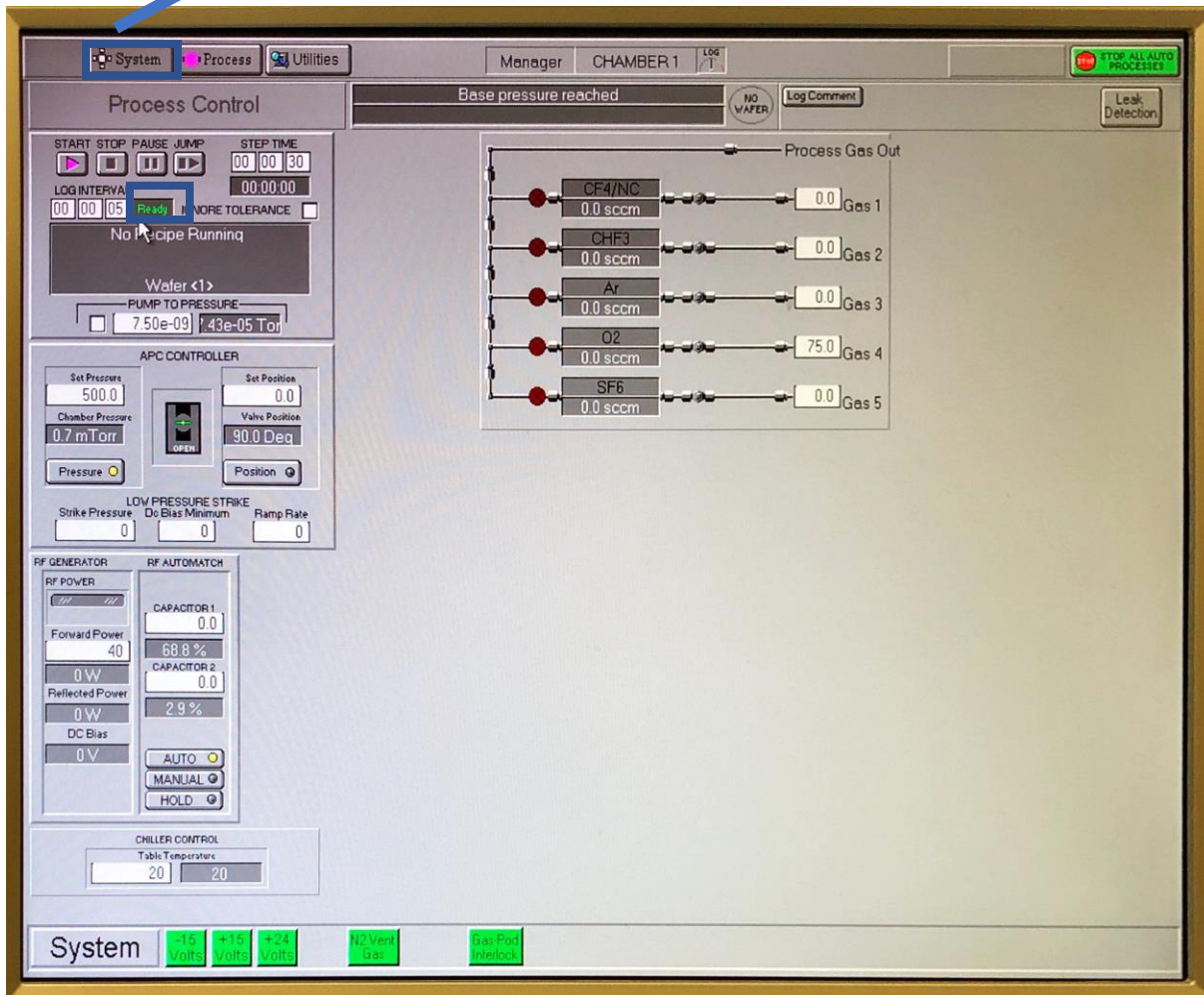


33. If you are running the process for a longer time, look out for the temperatures of ACT and ATP. If the ACT is showing high temp, be cautious and check for any warnings on the computer screen



34. After the process is finished, ACCEPT ALL THE YELLOW ALERTS and wait for the PROCESS CONTROL to show READY.

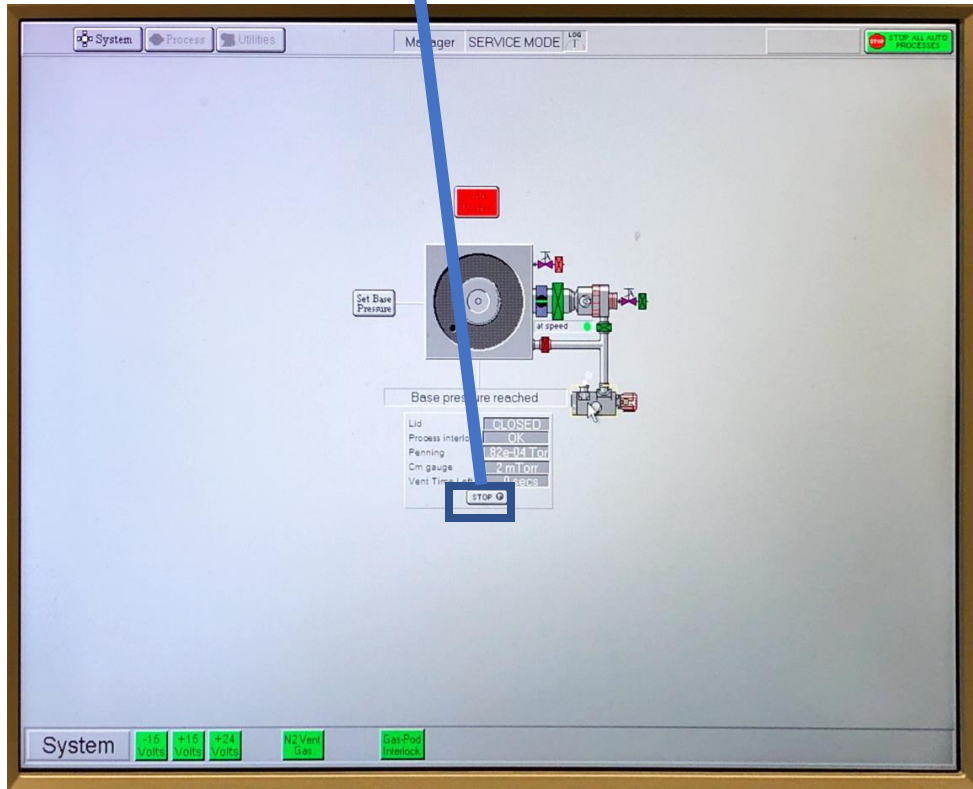
35. Go to "System" and choose "Pumping"



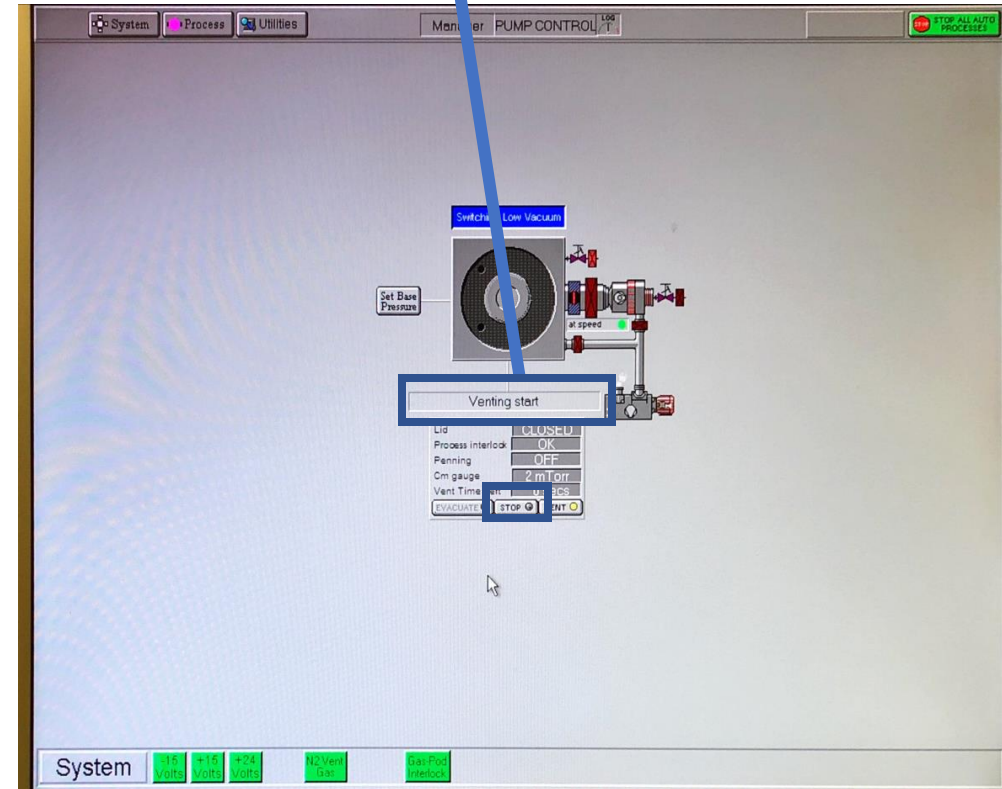
36. Turn off "AC LINE"... SLOWLY



37. "Stop" and then "Vent"



38. "Stop" the venting when it shows "Finished"

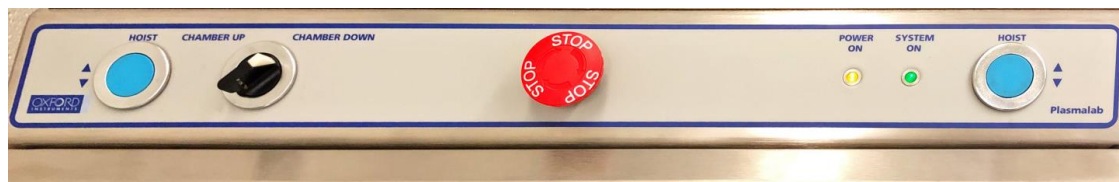
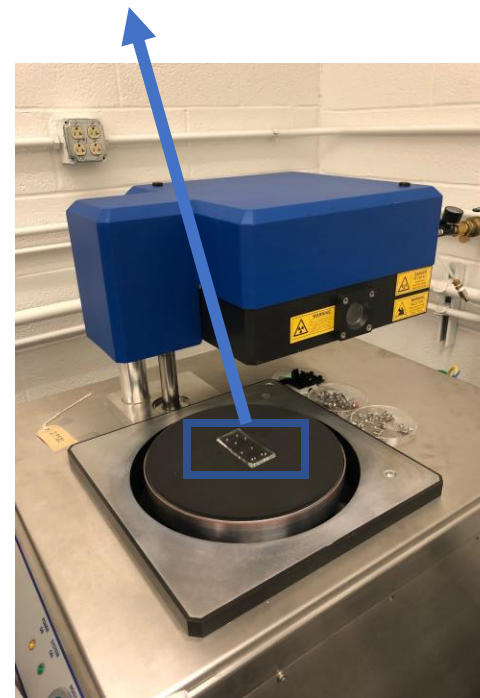


39. switch to "CHAMBER UP"



40. Press both button until the lid totally opens

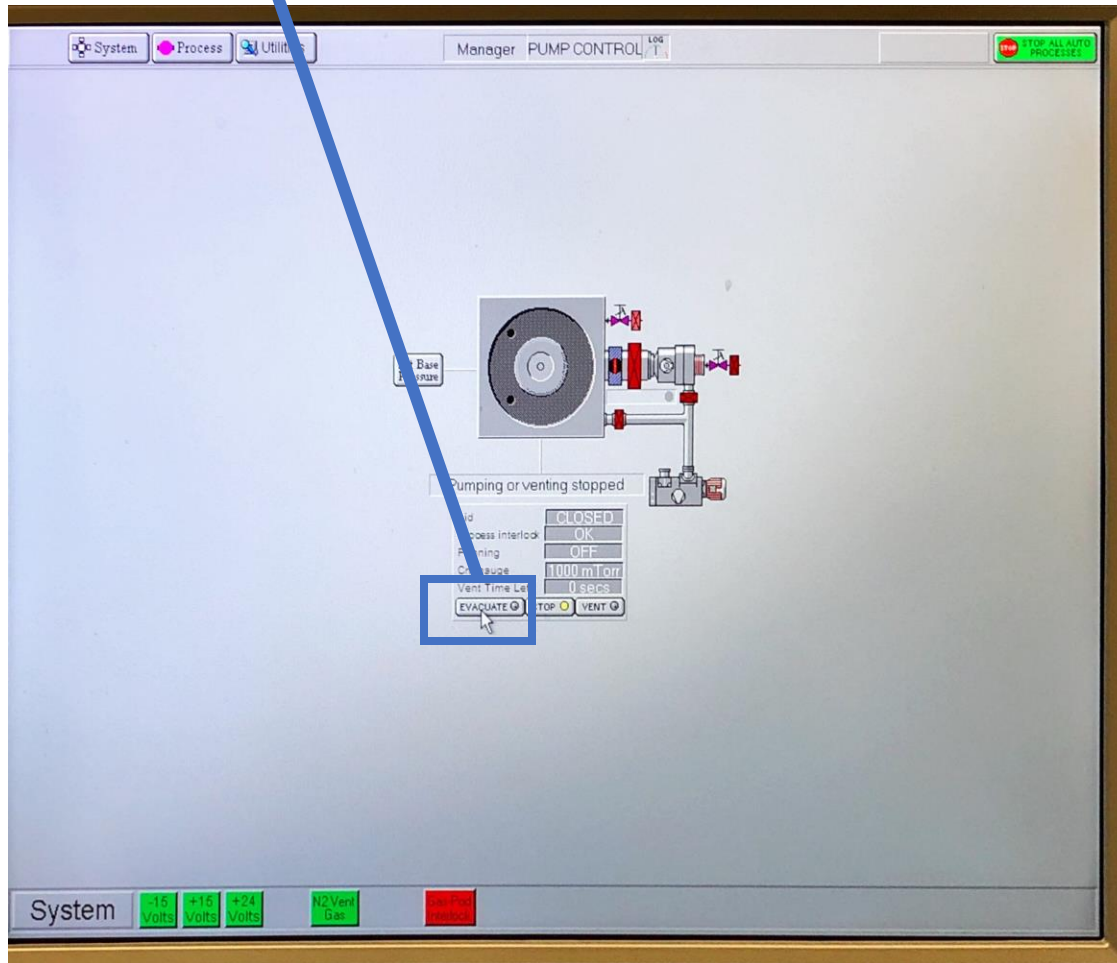
41. collect sample away from the stage (or leave it in if needed more time)



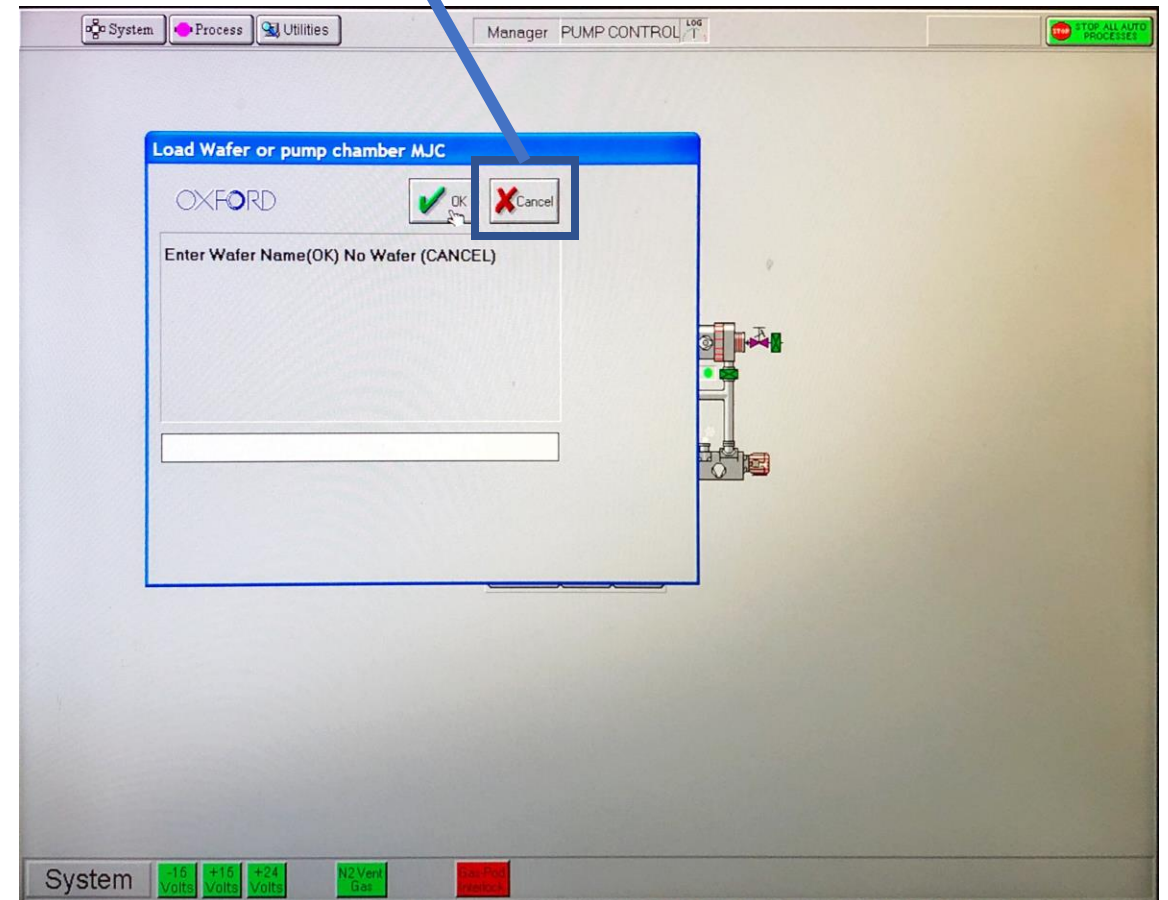
42. Switch to "CHAMBER DOWN" and press both hoists until the lid is completely closed.

- Repeat the steps from 20 to 42 if you are repeating the process.
- If you are done, follow the next steps to shut the machine down.

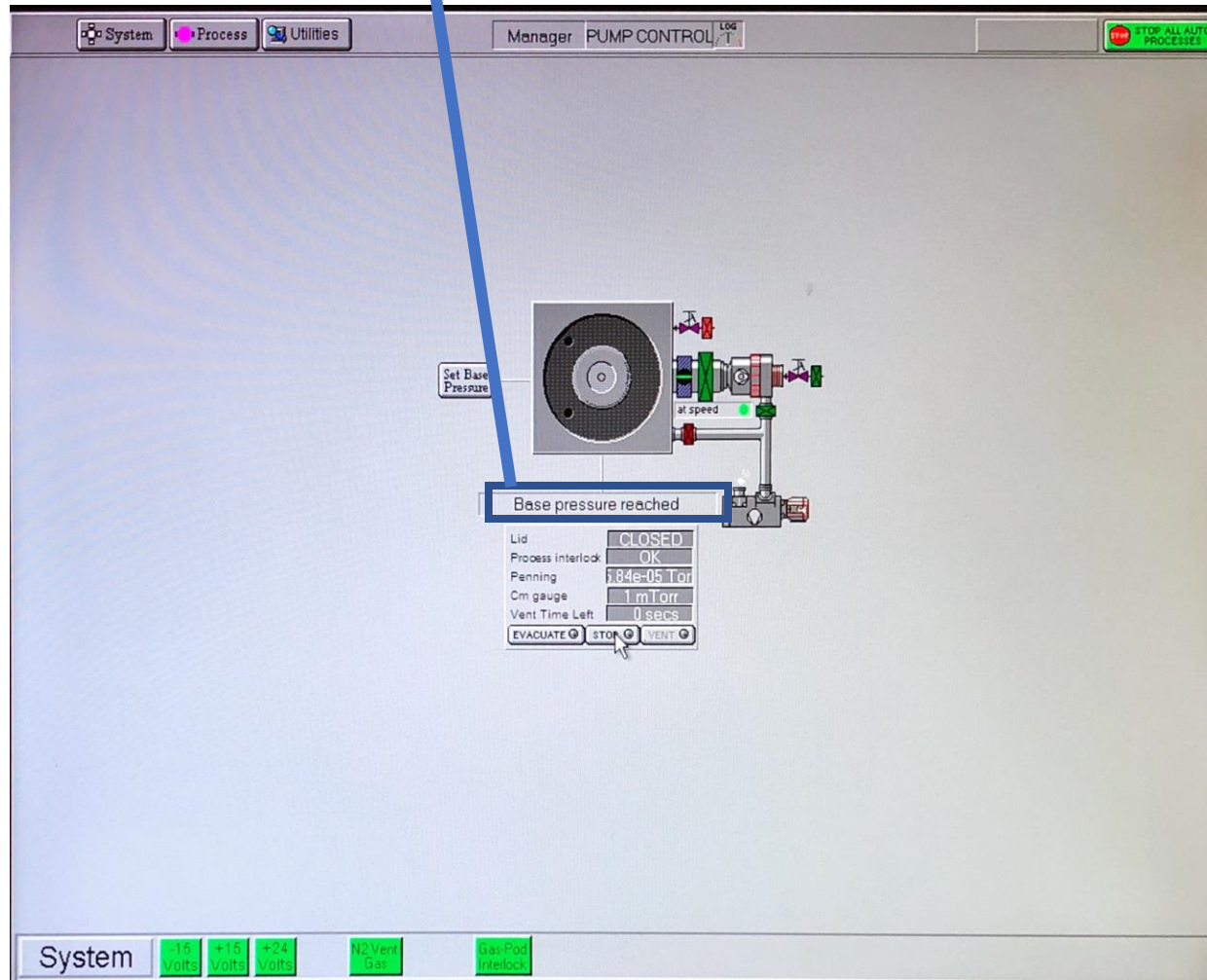
43. Click "Evacuate"



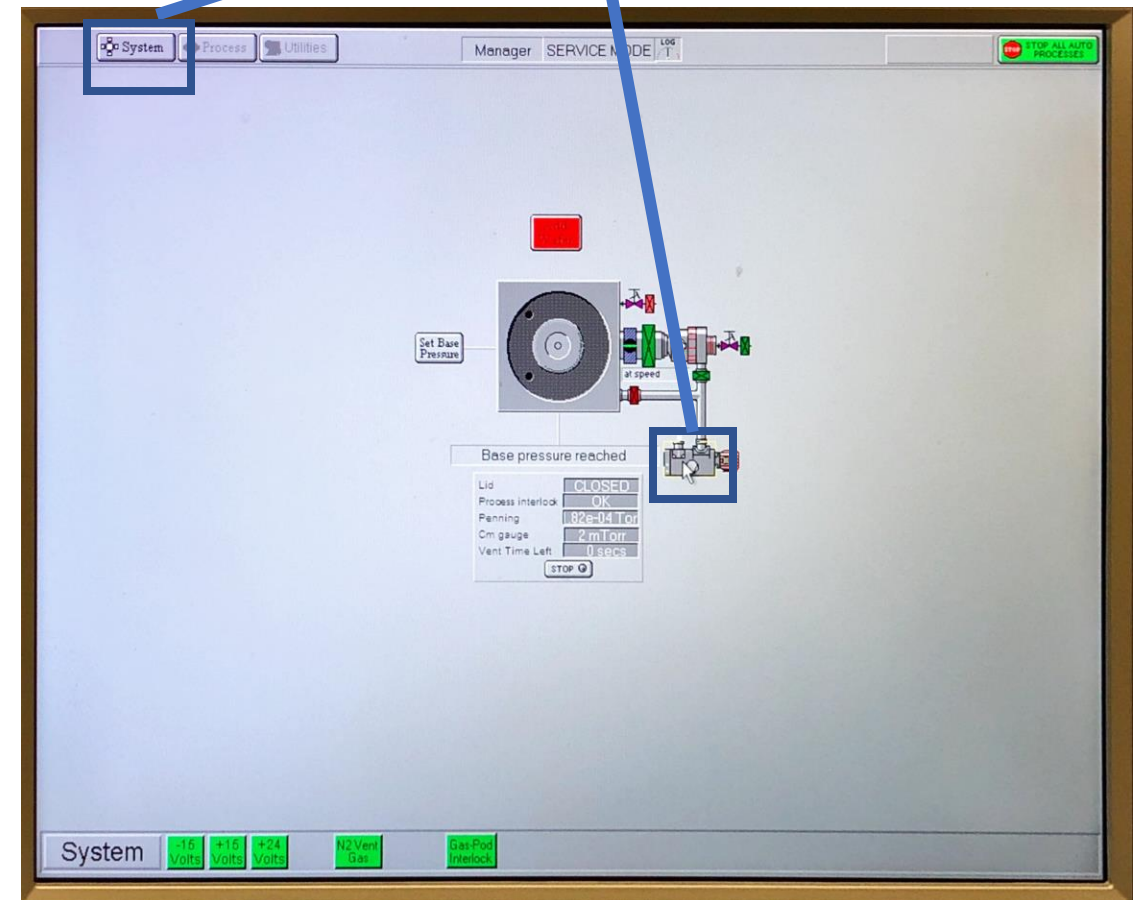
44. Click "Cancel" as there is no wafer in it



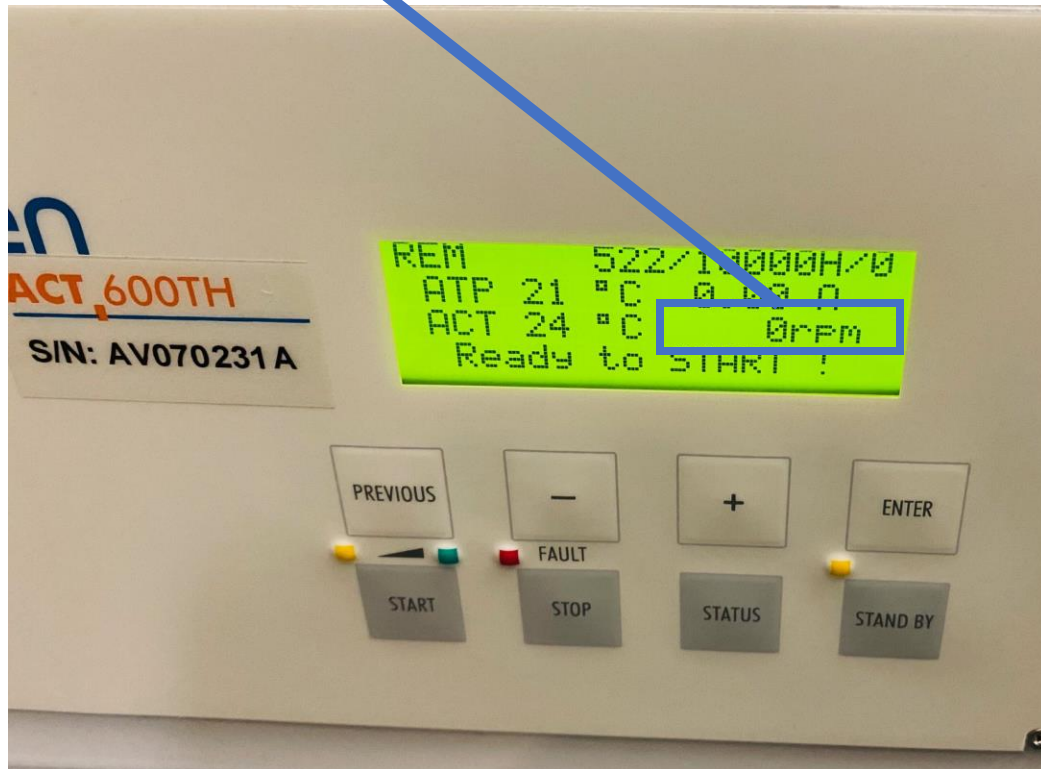
45. Wait for the screen to show "Base pressure reached"



46. Go to "System" - "Service" then click the "Turbo pump" image and click YES to turn it off

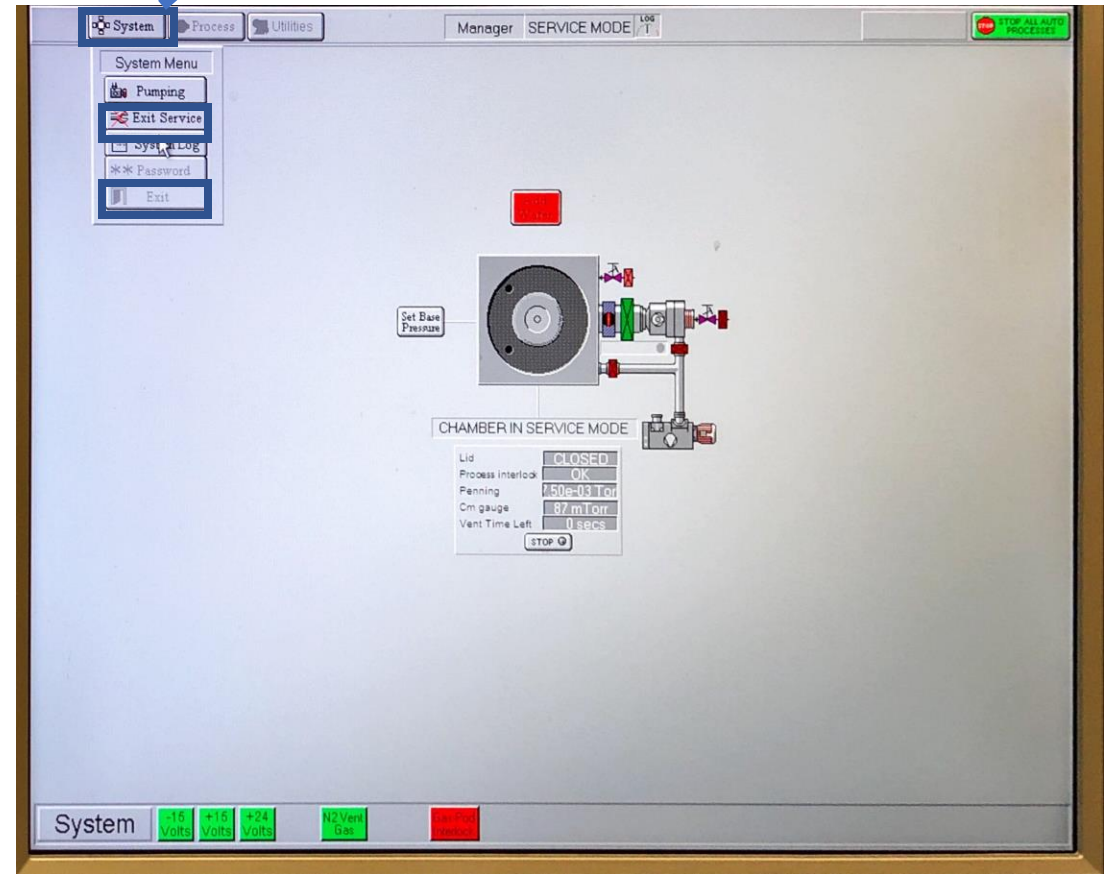


47. wait for the turbo to drop 0 rpm

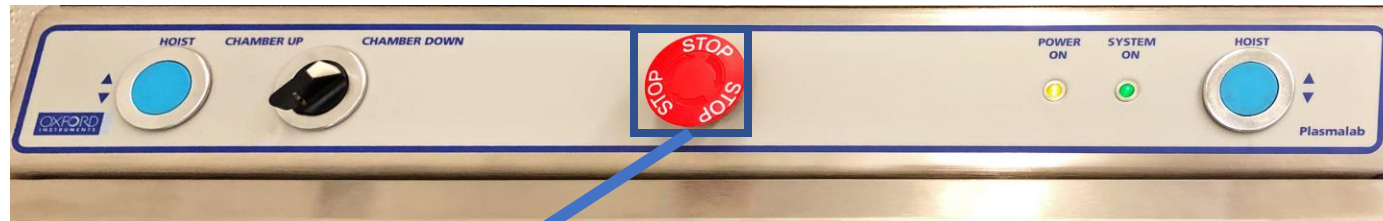


48. go to "System" - "Exit Service" and click YES to exit the service

49. Go to "System" - "Exit System" to exit the software



50. Push the red button to turn off power, it also turn off the small chiller

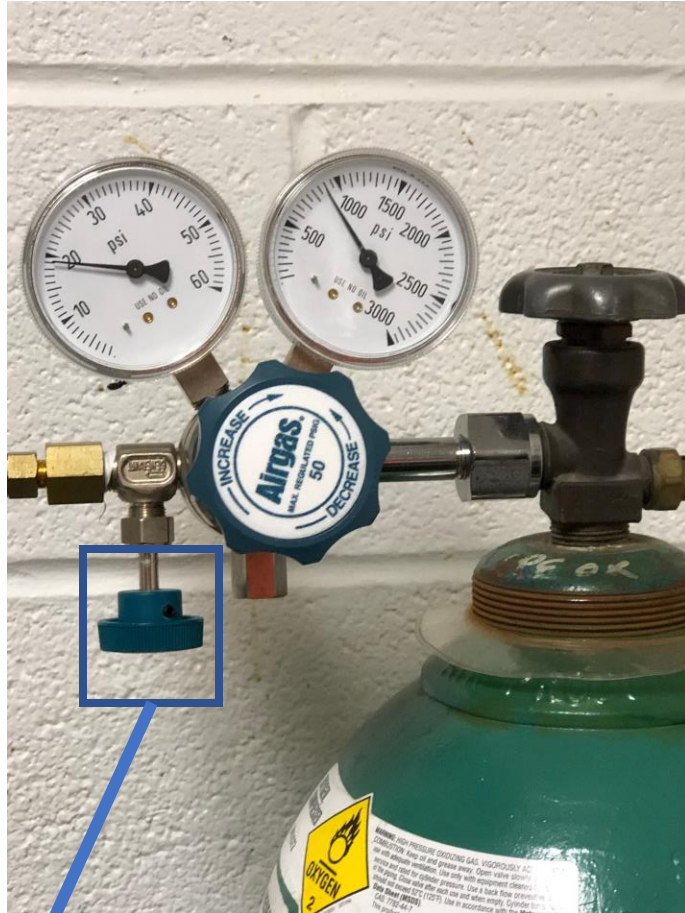


51. press "STOP" button



52. turn off both Compressor and N2 Valves (picture showing is on)

Back room



53. rotate the Valve to the right to close it



54. turn off "COOLING"

55. turn off "MAINS"